

An Adaptive Receiver Search Strategy for Electronic Support

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Abstract—In this paper, a search strategy for the detection of multifunctional radar emitters is proposed. The majority of today's literature regarding this topic models the intercept problem as that of the coincidence of two or more periodic window functions. Since this model is rather simplistic, in this paper the radars' illumination patterns are described by signal-to-noise ratio time series. To optimise the search dwells executed by an electronic support receiver, a constantly updated scan period estimate is calculated using the autocorrelation. The performance of the proposed algorithm is shown in comparison to several other approaches using a simulation framework.

I. INTRODUCTION

Intercepting the radiations of potentially threatening emitters, which usually are radars, is a common task in the field of electronic support (ES). Radars of interest might be operating in a wide range of frequencies, spanning tens of gigahertz. An intercept receiver covering the complete frequency range at the same time with a high sensitivity would be expensive, large and heavy. To overcome this problem, the frequency range of interest is usually divided into smaller bands of the receiver's instantaneous bandwidth and scanned sequentially. A schedule which defines how long and when to dwell on which band constitutes a search strategy. Normally, the emitters to be intercepted use a directional antenna and scan their surveillance region. Thus, an intercept is only possible when the receiver is tuned to the frequency of the emitter and the emitter's main beam or a sidelobe is directed towards the receiver.

In the literature, this situation is commonly modelled using two window functions, which are periodically "on" and "off" (see e.g. [1]). One of the window functions describes the points in time when the emitter illuminates the receiver and the other one specifies the events of the (omni-directional) receiver being tuned to the frequency of the emitter. When both window functions are "on" at the same time, an intercept occurs.

Modelling the illuminations by a strictly periodic window function simplifies the situation significantly. However, this simplification is not representative of modern multifunction radars. Today's radars perform searching and tracking - and sometimes even other functions like communication and missile guidance - in an interleaved manner, breaking the periodicity of the scan pattern. Moreover, electronically steered phased array antennas are capable of instantaneously

steering their beams and are not limited by mechanical parts which need to be rotated. Many authors (e.g. [2]–[8]) don't mention this fact or explicitly state that they don't consider phased array antennas (e.g. [9]). In [10], the authors argue that "due to their scanning function electronic beam steering antennas produce roughly periodic illuminations". Thus, they add jitter to the scan period, but still use a window function based model. In this paper, a complete antenna pattern model of an electronically steered phased array is used, allowing the sequence of illuminations to be described by a signal-to-noise ratio (SNR) time series. In addition, window functions usually only consider the main beam of the radar. In the presented approach, the radars can be intercepted and detected through the sidelobes as well.

Section II describes the developed algorithm in detail. The method used for evaluation and the results obtained are presented in Section III. Section IV gives the conclusion.

II. DEVELOPED ALGORITHM

In the presented algorithm, search and tracking dwells executed by the radars are considered. It is assumed that the receiver can differentiate between the two types of dwells, which is not an unrealistic assumption since typically different waveforms are used for different tasks.

At system startup, no information is available yet so the receiver begins scanning the frequency range of interest randomly. It uses a fixed dwell time and a fixed instantaneous bandwidth. SNR is calculated as given in Eq. 1. Here, G_t represents the emitter's antenna pattern. For the omni-directional receiver, G_r is assumed constant, independent from the angle of arrival (AoA). Note that this is a model of the SNR sufficient for our purposes, not necessarily reflecting reality exactly.

$$SNR = \frac{\left(\sum_{p \in P} \frac{PK(p) \cdot PW(p)}{PRI(p)} \right) \cdot \lambda^2 \cdot G_t \cdot G_r}{(4\pi R)^2 \cdot k \cdot T \cdot B \cdot L_t \cdot L_r} \quad (1)$$

p : pulse received

P : set of pulses received

$PK(p)$: peak power of pulse p

$PW(p)$: pulse width of pulse p

$PRI(p)$: pulse repetition interval of pulse p
 λ : wavelength of the pulses received
 G_t : transmit antenna gain in the direction of the receiver
 G_r : receiver antenna gain
 R : range between receiver and emitter
 k : Boltzmann constant
 T : noise temperature
 B : receiver's instantaneous bandwidth
 L_t, L_r : transmit and receive losses

Once the SNR on a band crosses the detection threshold T_D - a parameter of the receiver - the corresponding radio frequency (RF) is added to a list of active, tentative frequencies and the receiver stays tuned to that RF band for another d dwells. If the last one of these dwells results in the highest SNR, it listens for another dwell. It does so in order to explore the times of maximum SNR as a high SNR is desirable for accurate measurements of parameters like pulse repetition interval (PRI) or pulse width.

Bands in the list of tentative frequencies are visited more often in the further operation of the receiver in order to gain more knowledge about the illumination pattern. Algorithm 1 shows the procedure for selecting the next band. The probability for choosing each tentative frequency is scaled by y according to the number of frequencies in the list of tentative RFs until the scaled value reaches a maximum z . This is done to avoid dwelling on just very few frequencies with a very high probability. Scaling parameter y and maximum value z must lie between zero and one. This part constitutes the exploration phase of the strategy.

Algorithm 1 Select the next band in exploration phase.

Input: List of all RF bands (RFs), list of tentative RFs ($tentativeRFs$), scaling parameter y , max. value z
Output: Band for the next receiver dwell.

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1:  $r \leftarrow \text{random} \in (0, 1)$ 
2: if  $r < \min(|tentativeRFs| \cdot y, z)$  then
3:   return random band  $\in RFs \setminus tentativeRFs$ 
4: else
5:   return random band  $\in tentativeRFs$ 
6: end if
  
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If the receiver intercepts a search dwell and the SNR crosses the detection threshold, the autocorrelation of the intercepted SNR time series on that band is calculated. The autocorrelation is defined as the cross-correlation of a signal with itself. In the case at hand, the SNR time series is a signal discretised by the receiver dwells. Thus, the autocorrelation is given by

$$(f \star f)(n) = \sum_{m=-\infty}^{\infty} f^*(m)f(m+n) \quad (2)$$

$n = 0, \pm 1, \pm 2, \dots$

where f^* is the complex conjugate of f with f being the intercepted SNR.

An estimate of the radar's scan period $\hat{p}$ is given by the position of the maximum of the autocorrelation, excluding the value at lag zero, multiplied by the receiver's dwell time t_{dwell} :

$$\hat{p} = \left| \operatorname{argmax}_{n \neq 0} \{ (f \star f)(n) \} \right| \cdot t_{dwell} \quad (3)$$

If the standard deviation of the last j period estimates is smaller than a threshold T_{std} , the period is declared stable and dwells are scheduled for that frequency. Moreover, the band is removed from the list of tentative RFs because the exploration phase is finished. As an intercept is desired to happen when the SNR is maximal, future dwells are planned at the time of the last recorded global SNR maximum SNR_{max} plus integer multiples of the estimated period. Since the period can only be approximate due to interleaving with tracking dwells and uncertainties in estimation, the period is updated with each new detection of the corresponding emitter and dwells are rescheduled when a new global maximum in SNR is found. To account for a shift in period, only the last k SNR values are considered. Furthermore, several dwells are executed around the expected time of the next (local) SNR maximum. When a scheduled visit of the frequency band results in a SNR smaller than the threshold

$$T_{snr} = x \cdot SNR_{max} \quad (4)$$

with x set to be between zero and one, the number of dwells scheduled for the next visit is increased by i . After s misdetections, the period estimate is reset, the scheduled dwells are deleted and the band is again added to the list of tentative RFs, thus restarting the exploration phase.

When a tracking dwell crosses the detection threshold, a histogram of the differences in time of arrival (TOA) of the pulse bursts emitted by the radar on the corresponding frequency band is built. The maximum of that histogram is used as an estimate of one of the revisit times. Note that it is assumed that distinguishing the tracking dwells for different targets is not possible. Thus, several hypotheses for revisit times are kept. Each estimated revisit time generates a schedule for visiting the corresponding band. As for the search case, the RF is deleted from the list of tentative RFs and several dwells are executed around the scheduled time. However, as the targets move, the tracking dwells might quickly point into a direction where an intercept is not possible anymore. In addition, the revisit time estimates are rather rough, so scheduled dwells and the corresponding period estimate are directly deleted after the first misdetection. If after this deletion no revisit time estimate exists anymore, the frequency is added again to the list of tentative RFs.

When there is no dwell scheduled, a random band is chosen with a higher probability for the frequencies in the list of tentative RFs (see Alg. 1).

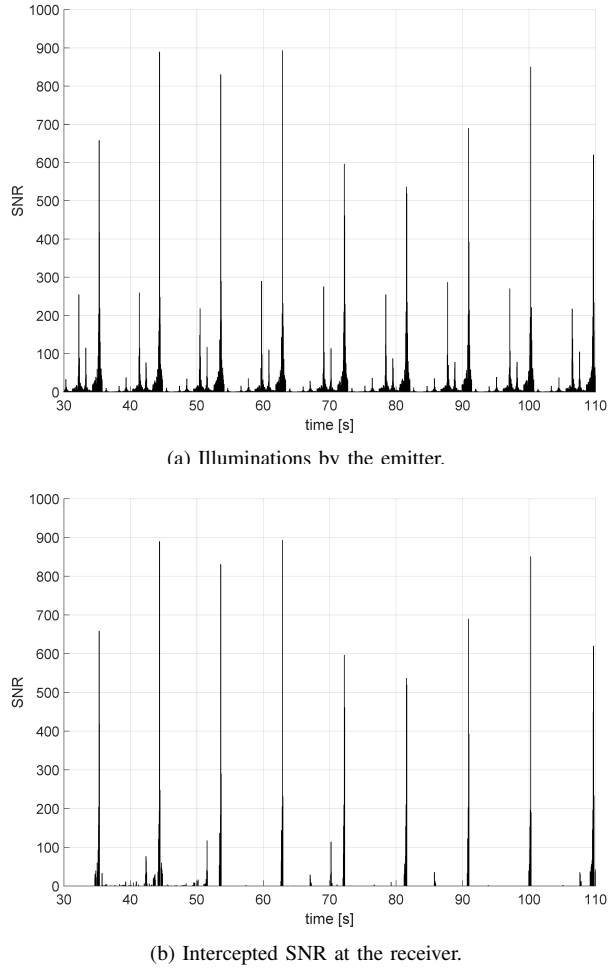


Fig. 1. Example of the illuminations generated by the emitter and the intercepted SNR time series at the receiver. The illumination pattern results from calculating the SNR at the receiver assuming that it's tuned the correct frequency the whole time.

Figure 1 shows an example of the illuminations generated by the emitter and the interceptions by the receiver. The SNR time series of Figure 1a depicts the illumination pattern that would result from the receiver being tuned to the frequency of the emitter for the whole time. In Figure 1b, the intercepted SNR at the receiver is shown. At (most of) the points in time where the SNR is zero, the receiver is tuned to a different frequency. From about 35 to 52 s simulated time many values are greater than zero, showing the exploration phase of the receiver in which it is tuned more often to the corresponding frequency. After the exploration phase is finished at about 52 seconds, it found a stable period estimate and starts scheduling for the local maxima of the SNR, as can be seen by the regularly spaced peaks. The dwells in between the SNR peaks are caused by the random search pattern used when there is no dwell scheduled. It also dwells on the known bands to be able to find a potentially undiscovered SNR maximum.

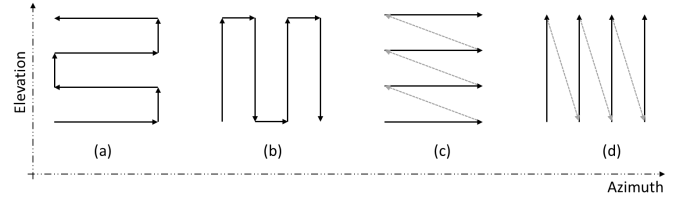


Fig. 2. Schematic representation of the scan patterns used. Grey arrows indicate a jump in beam position without dwelling.

III. EVALUATION & RESULTS

In the following, the simulation framework used for evaluation as well as the results are presented.

A. Simulation Framework

Evaluation of the algorithm is performed using a simulation framework. An airborne - but for the sake of simplicity stationary - receiver tries to intercept and detect ground-based radars using phased array antennas. The stationary emitters are placed randomly in the scenario, but it is made sure that they illuminate the receiver at some point. The faces of the radars are fixed, tilted backwards a few degrees for a better surveillance of the flying targets. Those are placed randomly as well and are assigned a random moving direction and speed. Based on the target's speed, a revisit time is calculated which is used by the radars for scheduling tracking dwells. Without any targets, the emitters perform scanning of their surveillance region with one of four different patterns (see Fig. 2). When a target is in a radar's field of view, tracking dwells are inserted into the scan pattern, which is then delayed. Consequently, there is no strict periodicity in the illumination patterns anymore. Moreover, it means that more tracked targets cause an increase in aperiodicity.

Each radar uses a different operating mode for searching and tracking. These are randomly assigned and consist of different frequencies, PRIs and pulse widths. There is only one radar transmitting on each frequency at the same time. This choice was made to avoid implementing deinterleaving algorithms. Nevertheless, if assumed that the radars can be deinterleaved based on their signal characteristics, also multiple radars on the same frequency can be allowed.

During evaluation, the developed algorithm is compared to a slightly different version of itself and two other algorithms. The considered strategies are:

- Adaptive strategy: The algorithm proposed in this paper as described above.
- Adaptive, no tracking: A version of the developed algorithm without scheduling for tracking dwells, included to see the impact of estimating the track revisit times of the radars.
- Active RFs: Starts with a random search, then uses the list of active, tentative RFs and dwells on these with a higher probability, like in the exploration phase of the presented algorithm.

- Random: Just selects a random band for each dwell, but makes sure that each band is selected about the same number of times.

In each simulation, the number of targets is varied whereas the simulated time is always 5 min. Further constant parameters are the instantaneous bandwidth of 250 MHz and the dwell time of 50 ms for each receiver. Moreover, there are always 10 radars present and the scanned frequency band is 2-18 GHz. Each simulation consists of 30 runs. In each run, the scenario is created randomly but the different receiver search strategies are executed in the same scenario before creating a new one. Note that each radar illuminates the receiver at some point, but the SNR values of the illuminations don't necessarily cross the detection threshold. Thus, the results shown in the next section are to be seen as a comparison and not an absolute performance measure, since reaching a 100% detection rate might not be possible.

B. Results

In Figures 3-5 the results for the simulations are presented. Each figure shows the values for one of these criteria:

- Efficiency (%): Percentage of dwells on bands with an active emitter which resulted in a detection.
- Total number of detections over all of the simulation runs.
- Detected (%): Percentage of radars that was detected at least once.

As the results show, the two versions of the adaptive strategy always perform better in terms of efficiency and total number of detections. The percentage of radars detected at least once is best for the random strategy. This is due to the fact that the random strategy only explores the environment without exploiting the information it obtains. Thus, its efficiency as well as total number of detections are always much worse.

Exploiting the knowledge about active frequencies clearly outperforms the random strategy in terms of efficiency and number of detections. Estimating the radars' scan periods and scheduling according to the estimates increases the efficiency and the number of detections even more. However, scheduling for the tracking dwells doesn't seem to make a major difference in performance. As mentioned earlier, the revisit time estimates are rather rough and as the targets move, at some time the tracking dwells point in a direction where a detection is not possible anymore. Moreover, as opposed to the search pattern with several illuminations as the emitter sweeps past the receiver's position, the tracking dwells are only detectable for a short time instance which requires an accurate schedule. Thus, because of misdetections for scheduled tracking dwells, the algorithm directly deletes the corresponding estimates.

None of the criteria considered seems to depend on the number of targets present in the simulation environment. This might change when much more targets are present such that the radars are overloaded with tracking and only perform the required minimum of searching. An evaluation of this situation is planned for the future.

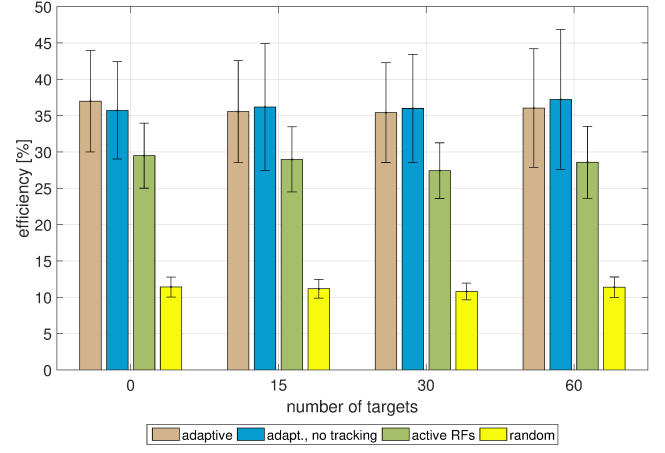


Fig. 3. Mean efficiency and standard deviation for the different search strategies and scenarios.

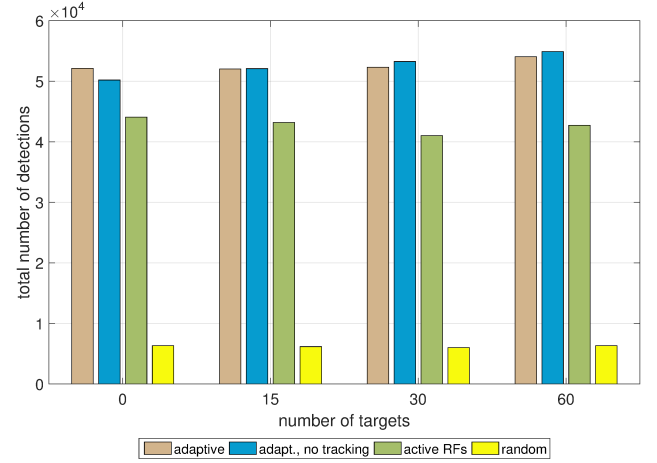


Fig. 4. Total number of detections for the different search strategies and scenarios.

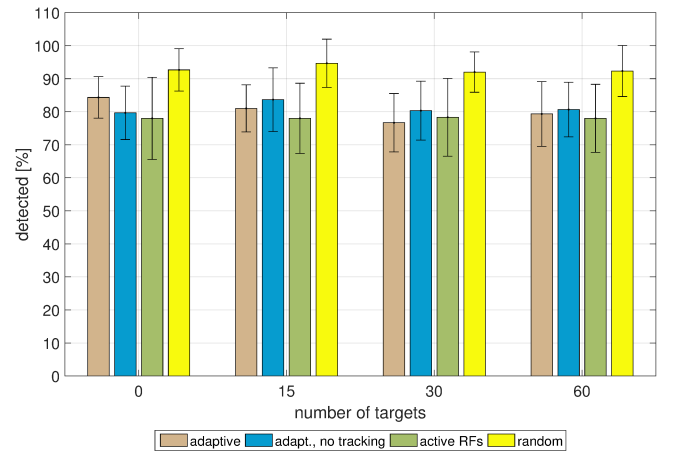


Fig. 5. Mean percentage of radars detected and standard deviation for the different search strategies and scenarios.

IV. CONCLUSION

An adaptive search strategy for the detection of multifunction radars has been proposed. The results obtained using simulations show that efficiency is improved significantly and more detections are achieved compared to non-adaptive algorithms. Also, modelling of the intercept/detection problem in electronic support using antenna patterns and not window functions was introduced. Using the intercepted SNR time series instead of a simple “on/off”-function allows for the detection of the radars’ sidelobes. In addition, a more sophisticated use of the information collected is possible. This includes scheduling for the times of maximum SNR for an accurate measurement of the radars’ waveform parameters and a more efficient use of the receiver’s limited resources. The simulations demonstrate that the less complex version of the search strategy, which doesn’t schedule for tracking dwells, provides an equally good performance as the complete algorithm. Thus, when regarding computational costs, the simpler version is a good approach.

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